Mengdie Huang (Maggie)

PostDoctoral Research Associate, Purdue University

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RESEARCH INTEREST

My research interests lie in security and privacy in machine learning, networking, and multimedia. I have been focusing on the robustness of deep neural networks and large (vision) language models against adversarial attacks and distribution shifts, with particular emphasis on robust training and certified defenses.
I have a strong background in deep learning, including techniques like manifold learning, randomized smoothing, transfer learning, contrastive learning, knowledge distillation, parameter-efficient fine-tuning, unlearning, and membership inference, applied to prediction and generation tasks in image, text, and network traffic domains.

EDUCATION

Purdue University, West Lafayette, IN — Sep 2022 – Jun 2025
Visiting Ph.D. in Computer Science
Advisor: Dr. Elisa Bertino

Xidian University, Xi'an, China — Sep 2019 – Jun 2025
Thesis: *Research on Key Techniques for Adversarial Robustness of Deep Neural Networks*.
Advisor: Dr. Xiaofeng Chen

Communication University of China, Beijing, China — Sep 2017 – Jun 2019
Master in Electronics and Communication Engineering
Bachelor in Radio and Television Editing

Huai'an University, Huai'an, China — Sep 2013 – Jun 2017
Bachelor in Communication Engineering

PUBLICATIONS

Papers Published

[1] **Mengdie Huang**, Yingjun Lin, Ninghui Li, Xiaofeng Chen, Elisa Bertino✉. TriXfer: Triple-Cross Transfer Attack on Vision Transformer–Adapted Downstream Classifiers[J]. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2026. (CCF A)

[2] **Mengdie Huang**, Yingjun Lin, Ninghui Li, Xiaofeng Chen, Elisa Bertino✉. CARD: Robustness-Preserving Transfer Learning for Network Intrusion Detection via Contrastive Adversarial Representation Distillation[J]. *IEEE Transactions on Dependable and Secure Computing (TDSC)*, 2025, pp. 5134-5151. (CCF A)

[3] **Mengdie Huang**, Yingjun Lin, Xiaofeng Chen, Elisa Bertino✉. Dimensional Robustness Certification for Deep Neural Networks in Network Intrusion Detection Systems[J]. *ACM Transactions on Privacy and Security (TOPS)*, 2025, pp. 1-33. (CCF B)

[4] **Mengdie Huang**, Yingjun Lin, Xiaofeng Chen✉, Elisa Bertino. MARS: Robustness Certification for Deep Network Intrusion Detectors via Multi-Order Adaptive Randomized Smoothing[C]. *IEEE International Conference on Trust, Security, and Privacy in Computing and Communications (TrustCom)*, 2024, pp. 767-774. (**Best Paper**) (CCF C)

[5] **Mengdie Huang**✉, Hyunwoo Lee, Ashish Kundu, Xiaofeng Chen, Anand Mudgerikar, Ninghui Li, Elisa Bertino. ARIoTEDef: Adversarially Robust IoT Early Defense System based on Self-evolution against Multi-step Attacks[J]. *ACM Transactions on Internet of Things (TIOT)*, 2024, 5(3): 1-34.

[6] **Mengdie Huang**, Yi Xie, Xiaofeng Chen✉, Jin Li, Changyu Dong, Zheli Liu, Willy Susilo. Boost Off/On-Manifold Adversarial Robustness for Deep Learning with Latent Representation Mixup[C]. *ACM Asia Confer-*

ence on Computer and Communications Security (AsiaCCS), 2023, pp. 716-730. (CCF C)

[7] **Mengdie Huang**, Cheng Yang✉, Hao Li, Jian Shen. Sparse Selective Encryption for HEVC 4K Video Using Spatial Error Spread[J]. *Journal of Internet Technology (JIT)*, 2019, 20(5): 1589-1600.

[8] **Mengdie Huang**, Cheng Yang✉, Yuan Zhang. Selective Encryption of H.264/AVC based on Block Weight Model[C]. *IEEE International Conference on Communication Technology (ICCT)*, 2018, pp. 1368-1373.

[9] Elisa Bertino✉, Hyunwoo Lee, **Mengdie Huang**, Charalampos Katsis, Zilin Shen, Bruno Ribeiro, Daniel De Mello, Ashish Kundu. A Pro-Active Defense Framework for IoT Systems[C]. *IEEE International Conference on Collaboration and Internet Computing (CIC)*, 2023, pp. 125-132.

[10] Yi Xie, **Mengdie Huang**, Xiaoyu Zhang, Changyu Dong, Willy Susilo, Xiaofeng Chen✉. GAME: Generative-based Adaptive Model Extraction Attack[C]. *European Symposium on Research in Computer Security (ESORICS)*, 2022, pp. 570-588. (CCF B)

[11] Shanyue Bu✉, **Mengdie Huang**, Kun Yu. An Effective Scheme for Provable Data Possession[C]. *International Conference on Intelligent Control and Computer Application (ICCA)*, 2016, pp. 344-348.

Papers Under Review

[1] Chen Peng*, **Mengdie Huang***✉, Ninghui Li, Elisa Bertino. Local Knowledge for Global Impact: Targeted Evasion Attack Against Vertical Federated Learning, 2026. (∗ joint first authors.)

[2] MD Shamsul Kaonain*, **Mengdie Huang***✉, Imtiaz Karim, Elisa Bertino. CARE: Robust Transformer-based Network Intrusion Detection via Consistency-aware Attention-Representation Enhancement, 2026. (∗ joint first authors)

[3] Mir Imtiaz Mostafiz*, **Mengdie Huang***✉, Imtiaz Karim, Elisa Bertino. GRAFT: Adversarially Robust Transfer Learning for Graphs Under Structural Data Scarcity, 2026. (∗ joint first authors)

[4] Yani Liu, Feng Zhang✉, Yu Zhang, **Mengdie Huang**, Elisa Bertino, Xiaoyong Du. EntroDelta: An Error-Bounded, Data-Free Compression Framework for Massive Model Storage. 2026.

[5] Zhongkui Ma, **Mengdie Huang**, Guangdong Bai✉, Elisa Bertino, Zihan Wang. Local Nonlinear Substitution for Bounding Activation Functions, 2026.

Books

[1] Xiaofeng Chen✉, **Mengdie Huang**, etc. Cloud Computing Security (2nd Edition)[B]. *Science Press. China Science Publishing & Media Co., Ltd.* 2023.

[2] Xiaofeng Chen✉, **Mengdie Huang**, etc. Searchable Encryption[B]. *Science Press. China Science Publishing & Media Co., Ltd.* 2024.

Employment

Purdue University, West Lafayette, IN — July 2025 – Present

PostDoctoral Research Associate in the Department of Computer Science

Advisor: Dr. Elisa Bertino and Dr. Ninghui Li

Research: Robustness of large language and vision foundation models.

Purdue University, West Lafayette, IN — Sep 2022 - June 2025

Visiting Scholar (PhD Student) in the Department of Computer Science

Advisor: Dr. Elisa Bertino

Research: Robustness of deep learning-based network traffic analysis systems.

Xidian University, Xi'an, China — Sep 2019 – Aug 2022

Research Assistant in the School of Cyber Engineering

Advisor: Dr. Xiaofeng Chen

Research: Robustness of deep learning-based image classification systems

Xidian University, Xi'an, China — Jul 2021 - Jul 2022

Teaching Assistant in the School of Cyber Engineering

Instructor: Dr. Miroslaw Kutylowski in *Distributed Computing*

Xidian University, Xi'an, China — Sep 2020 - Jan 2021

Teaching Assistant in the School of Cyber Engineering

Instructor: Dr. Jianfeng Wang in *Probability Theory and Mathematical Statistics*

National Radio and Television Administration, Beijing, China — May 2019 - Jul 2019

Research Assistant in *Academy of Broadcasting Science*

Research: Secure video coding standards.

ByteDance Ltd, Beijing, China — Aug 2018 - Sep 2018

Internship in *Commercialization Department*

GRANT WRITING EXPERIENCE

Core Author of the NSF proposal, Purdue – West Lafayette, IN — Sep 2026

Model Robustness, Ownership, and Privacy Preserving in Transfer learning

Author of the Amazon research proposal, Purdue – West Lafayette, IN — May 2026

Formal and Drift-Aware Containment of Agents in Regulated Systems

Core Author of the Cisco research proposal, Purdue – West Lafayette, IN — Mar 2023

Detection of GenAI Generated Malware Variants and Sandbox Evasion using GenAI,

Awarded $\$150,000$ by Cisco Research

ACADEMIC SERVICE

Journal Reviewer

- IEEE Transactions on Dependable and Secure Computing (TDSC): 2025 - Present
- IEEE Transactions on Networking (TNET): 2025 - Present
- IEEE Transactions on Parallel and Distributed Systems (TPDS): 2025-Present
- IEEE Transactions on Information Forensics & Security (TIFS): 2024 - Present
- IEEE Transactions on Neural Networks and Learning Systems (TNNLS): 2024 - Present
- ACM Transactions on Multimedia Computing, Communications, and Applications (TOMM): 2024 - Present
- Computers and Electrical Engineering (CEE): 2024 - Present
- Computing and Informatics (CAI): 2024 - Present
- American Journal of Artificial Intelligence (AJAI): 2024 - Present
- ACM Computing Surveys: 2023 - Present
- IEEE Transactions on Knowledge and Data Engineering (TKDE): 2022 - Present
- Telecommunication Systems (TELS): 2021 - Present
- Computer Standards & Interfaces (CSI): 2021 - Present
- Connection Science: 2020 - Present
- IEEE Transactions on Circuits and Systems for Video Technology (TCSVT): 2019 - Present

PC Member

- IEEE International Conference on Parallel and Distributed Systems (ICPADS): 2026
  - Track: Agentic Design for System and Network

Conference Reviewer

- International Conference on Data Security and Privacy Protection (DSPP): 2024, 2025
- Annual Computer Security Applications Conference (ACSAC): 2024

- European Symposium on Research in Computer Security (ESORICS): 2021, 2024, 2025
- ACM Conference on Data and Application Security and Privacy (CODASPY): 2023, 2024, 2025
- International Conference on Machine Learning for Cyber Security (ML4CS): 2022
- Information Security Conference (ISC): 2022
- International Conference on Information and Communications Security (ICICS): 2021

Conference SubReviewer

- IEEE Symposium on Security and Privacy (S&P): 2025, 2026
- ACM Conference on Computer and Communications Security (CCS): 2025
- USENIX Security (USENIX Security Symposium): 2026

Artifact Evaluation Committee Member

- ACM Conference on Computer and Communications Security (CCS): 2025, 2026
- Network and Distributed System Security Symposium (NDSS): 2026

PROJECTS

Pro-Active Attack Management for Edge Computing Security, West Lafayette, IN — Nov 2022 - Dec 2023

Developed a robust LSTM-based network intrusion detection system to defend against cyber chain attacks.

Sponsor: Cisco Systems, Inc.

Universal and Efficient 4K Video Protection Technology Supporting Coding Standard Extension, Beijing, China — Jun 2018 - Jun 2019

Developed A Secure and Efficient 4K HEVC Video Encoding Framework.

Sponsor: National College Students Innovation and Entrepreneurship Training Program.

MENTORING

Chen Peng, Ph.D. Student, Purdue University — Feb 2026 - Present

Research: Machine learning security.

Md Shamsul Kaonain, Ph.D. Student, Purdue University — Jun 2025 - Present

Research: Network Traffic Analysis with deep learning.

Yani Liu, Visiting Ph.D. Student, Purdue University — Mar 2025 - Present

Research: Data compression in machine learning.

Mir Imtiaz Mostafiz, Ph.D. Student, Purdue University — Jan 2025 - Present

Research: Robustness of graph neural networks.

Jason Hu, Undergraduate Student, Purdue University — Feb 2024 - Present

Research: Robust Transfer Learning.

Yingjun Lin, Undergraduate Student, Purdue University — Nov 2022 - Dec 2023

Research: Adversarial attack and robustness in deep learning.

Weihao Liu, Undergraduate student, Xi'an University of Posts and Telecommunications — Dec 2020 - Jun 2021

Research: Iterative adversarial attack with projected gradient descent.

Mengchen Wang, Undergraduate student, Xi'an University of Posts and Telecommunications — Dec 2020 - Jun 2021

Research: Adversarial attack with Jacobian-based Saliency Map.

Wei Li, Undergraduate student, Xidian University — Dec 2019 - Jun 2020

Research: Adversarial attack with Fast Gradient Sign Method.

Fengdong Li, Undergraduate student, Communication University of China — Sep 2018 - Jun 2019

Research: Selective Encryption for HEVC Video.

Yujie Wang, Undergraduate student, Communication University of China | Sep 2018 - Jun 2019

Research: Selective Encryption for HEVC Video.

TECHNOLOGY CONTEST AWARDS

Blue Bridge Cup National Software Development Technology Competition | Mar 2016

Jiangsu Province Second Prize, China

National College Student Innovation and Entrepreneurship Competition | Sep 2015

Jiangsu Province Third Prize, China

Challenge Cup National College Students Extracurricular & Academic Contest | Jun 2015

Jiangsu Province First Prize, China

SKILLS

Programming: Python, C++, HTML, Java, SQL, MATLAB
Tools: Pytorch, Sklearn, Keras, HuggingFace, VScode, etc.
Language: English, Chinese
Certificate: National Computer Rank Examination Grade 4 (NCRE-4), Network Engineer

SELECTED AWARDS & HONORS

- Outstanding Doctoral Student Awarded by Xidian University | Sep 2019 - Sep 2022
- Xidian University Doctoral Academic Scholarship for Four Consecutive Years | Sep 2019 - Sep 2022
- Merit Graduate Student Awarded by Communication University of China | Jun 2019
- Communication University of China Graduate Academic Scholarship | Sep 2017 - Sep 2018
- Integrated Communication Social Scholarship Supported by Ze Media | Sep 2018

REFERENCES

Dr. Elisa Bertino, Department of Computer Science, West Lafayette, IN | Purdue University
Dr. Ninghui Li, Department of Computer Science, West Lafayette, IN | Purdue University
Dr. Xiaofeng Chen, School of Cyber Engineering, Xi'an China | Xidian University